Karthik Jayaraman
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Experienced Software QA Automation Engineer proficient in various programming languages, such as C#, Typescript/JavaScript, Java, and Python. I enjoy all aspects of the testing process, from test strategy and test case creation to execution and defect tracking.

Detail-oriented problem-solver who is comfortable working in both structured and flexible development environments. Always looking for ways to improve efficiency and productivity. Proven ability to deliver high-quality software solutions through efficient test execution and defect resolution.

Skills

Programming Languages: C#, Typescript/JavaScript, Python,, Java

UI Automation Test Frameworks:
Spec-flow/MsTest Framework with C# and Selenium Web-driver, Appium
Cucumber test frameworks with Typescript/JavaScript and WebdriverIO
Cucumber framework with Java + Selenium webdriver
Playwright with Typescript
Cypress with Typescript

API Automation Test Framework: Spec-flow/MsTest Framework with C# and RestSharp

Testing and Versrion control tools: Azure DevOps, JIRA, Github and TFS

Cloud Services: Browserstack**,** LambdaTest, Sauce Labs, AWS (Lambda, EC2, SQS), Azure Pipeline, CircleCI

Databases: SQL Server, DynamoDb, MangoDb, Oracle

Profesional Summary

- Create an all-encompassing Test Strategy document that harmonizes with business needs and use cases. This document will outline the testing scope, test data setup, detail the automation approach, prioritize test scenarios according to risk assessment, and integrate pertinent scenarios into the regression suite.
- Proficient in crafting automation scripts for a wide range of end-to-end and front-end user interface testing scenarios using various testing frameworks over the years. This includes leveraging J**asmine/Cucumber test frameworks with Typescript/JavaScript** for Angular projects, as well as employing **WebdriverIO/Protractor/Selenium Web-driver.** Additionally, for C# projects, I have extensive experience with **Spec-flow, C#, and Selenium Web-driver**.
- Strong experience in using **playwright** and **cypress** for e2e UI automation scripts.
- Proficient in developing automation scripts for integration test scenarios, particularly for back-end/API web services, utilizing **C#, Rest-sharp, and the Spec-flow/X-Unit test framework**
- Skilled in crafting automation scripts for performance and load testing, both for API routes and web pages, using **JMeter and Microsoft's performance testing tools**.
- Extensive background in authoring automation test scripts for both hybrid and native mobile apps, leveraging **Typescript/C# and utilizing Appium** along with Appium Inspector to locate and interact with app elements.
- Proficient in maintaining and enhancing AWS Lambda functions written in Python for over three years.
- Developing and managing **CircleCI** YAML scripts for seamless integration of end-to-end flows within the CI/CD pipeline.
- Proficient in using **BrowserStack, Sauce Labs, and LambdaTest** for cross-browser and cross-platform testing.
- Executing User Interface test regression suites on Selenium Grid across multiple browsers, including Windows (IE and Chrome) and macOS (Chrome).
- Proficient in version control tools like **Github, TFS and ALM**
- Strong ability to work on **AWS Services like ECS, ECR, EC2, Lambda, DynamoDB and SQS.**
- Proficient in utilizing both GitHub and TFS for code maintenance, review, and deployment.
- Strong experience in **Backend testing, validating data against database, data migration and ETL testing**
- Strong experience in analyzing ROI on automating test scenarios.
- Excellent knowledge in RDBMS concepts and written complex **SQL queries**.
- Good knowledge in executing UNIX scripts and debugs issues.
- Strong ability to work on multiple projects and enhancements simultaneously.
- Provided mentorship to fellow team members on aspects such as test strategy, automation methodologies, and script development. Led and supervised testing teams.
- Experienced with coordinating cross-functional teams.

Work Experience

UNOS, Richmond, VA, Jan2024 - Present
Senior Test Automation Engineer

UNet:
Waitlist is the Web Application that is used to Add Candidates to the Waiting List to potentially get the gift of life! Waitlist refers to adding, changing or removing candidates from the National Transplant Candidate Waiting List.
DonorNet is the Web Application that is used to Add Donors, and from these Donors the Match can be ran.
TransNet:
This application is used to track various organs that are shipped from organ center to transplant center. It is responsible for verifying donor information, creating shipping label and confirming transplant center received expected organs.
Mobile app : Handing donor information and creating shipping label, works offline
Web app : Confirming expected organs and reshipping organs if needed

Responsibilities:

- Create Test case document based on business requirements which includes scope of testing, automation strategy, prioritize test scenarios based on risk assessment and adding scenarios to regression suite.
- Created behavior driven development framework for UI Automation from scratch using SpecFlow - MSTest, C# and Selenium Webdriver using page object pattern.
- Create and maintain Rest API automation scripts using C#, Rest Sharp, Entity Framework and Specflow
- Create and maintain test data generation tool using JMeter scripts.
- Create scripts to update test execution in Azure DevOps from Automation suite test results.
- Mentor testing team member on test case and automation design.
- Creating test cases and test execution using Azure DevOps.
- Reporting and tracking defects to deliver quality product.
- Creating task and estimating effort for assigned tasks on version one.
- Review backlog and part of creating epic and stories.
- Provide feedback on sprint progress in retrospective.
- Proficient in cross browser testing.

Environment: Selenium Webdriver, Selenium Grid, SQL Server, C#, Specflow, Rest API, Rest Sharp, Entity Framework, Appium, Team Foundation Server, Version one, Trello, Visual Studio, Page Object Pattern, C# Web development MVC pattern.

Staff Quality Engineer
Snagajob, Richmond, VA, Feb2020 – Oct2023

Snagajob is an online marketplace for hourly jobs and shifts. Helping employers to find great hourly talent worker and easy job finder for job seekers to find part-time and full-time jobs.

Responsibilities:

- Create Test case document based on business requirements which includes scope of testing, automation strategy, prioritize test scenarios based on risk assessment and adding scenarios to regression suite.
- Creating and maintaining automation scripts written in Typescript, WebdriverIO, Protractor, Jasmine and Cucumber (BDD).
- Implemented continuous integration through AWS parameter store which runs UI and API regression suite on every code commit and published report in regression application.
- Performance testing using JMeter.
- Create scripts to update test execution in JIRA from Automation suite test results.
- Mentor testing team member on test case and automation design.
- Reporting and tracking defects to deliver quality product.
- Creating task and estimating effort for assigned tasks on JIRA.
- Code maintenance with GITHUB.
- Review backlog and part of creating epic and stories.
- Provide feedback on sprint progress in retrospective.
- Testing microservices.
- Implemented proof of concept with cypress and compared its advantages and disadvantages with webdriverIO for e2e Automation.
- Create and maintain Rest API automation scripts using C#, Rest Sharp and Specflow.
- Proficient in cross browser testing.
- Created behavior driven development framework for UI Automation from scratch using Cucumbert, Typescript and WebdriverIO using page object pattern.
- Running UI test regression suite on AWS containers for various browsers.
- Experienced in mobile test automation using Appium and Typescript.
- Experienced in migration testing (ETL) moving data from old architect RDBMS to new architect.
- Heavy use of SQL in Data Migration testing.

Environment: WebdriverIO, Protractor, Angular, Typescript, Jasmine, Cucumber, C#, Specflow, Rest API, Rest Sharp, Appium, Github, JIRA, VS Code, Visual Studio, Page Object Pattern, API microservices and CircleCI.

UNOS, Richmond, VA, Dec2013 - Jan2020
Senior Test Automation Engineer

Responsibilities:

- Create Test case document based on business requirements which includes scope of testing, automation strategy, prioritize test scenarios based on risk assessment and adding scenarios to regression suite.
- Creating and maintaining automation scripts written in C#, Selenium Webdriver and Specflow (BDD).
- Mentor testing team member on test case and automation design.
- Creating test cases and test execution using team foundation server.
- Create test data generation tool using Microsoft Web performance and load testing tool.
- Reporting and tracking defects to deliver quality product.
- Creating task and estimating effort for assigned tasks on version one.
- Review backlog and part of creating epic and stories.
- Provide feedback on sprint progress in retrospective.
- Heavy use of SQL in Data Migration testing.
- Create and maintain Rest API automation scripts using C#, Rest Sharp, Entity Framework and Specflow
- Proficient in cross browser testing.
- Created behavior driven development framework for UI Automation from scratch using SpecFlow - MSTest, C# and Selenium Webdriver using page object pattern.
- Running UI test regression suite on Selenium Grid for various browsers (Win - IE and Chrome, Mac – Chrome).
- Experienced in mobile test automation using Appium and C#
- Fixed defects found during testing (C# and Web Forms).
- Developed a page in patient safety project in MVC C# pattern.
- Part of web development in DonorNet Mobile project – Angular 2.
- Developed a Rest API in DonorNet Mobile project.

Environment: Selenium Webdriver, Selenium Grid, SQL Server, C#, Specflow, Rest API, Rest Sharp, Entity Framework, Appium, Team Foundation Server, Version one, Trello, Visual Studio, Page Object Pattern, C# Web development MVC pattern.

Infosys, Sep2003 - Dec2013
QA Test Lead

I worked with below clients through Infosys,

Capital One Financial, Richmond, VA (Aug'08 - Dec'13)

Migration Factory (ETL Testing): Migration Factory is built using Xenos Developer Studio to migrate all documents and images from CMOD (IBM Content Manager OnDemand) version X to Y. More than billions of document and images are transferred in production. This project involves validating Migration Factory able to extract, convert and load all documents properly into target system.

SIMS Automation: SIMS is web based application developed in Java. This project is to automate regression test cases. Using HP Quick Test Professional automation tool, regression test cases are automated with keyword and data driven framework.

EI Web Service Regression Testing: EI Web service is used to getContent, addContent, getMetadata and updateMetadata. Web Service operation regression test cases are automated using SoapUI.

ACBS (Advanced Commercial Banking System): ACBS is loan servicing system integrated with many interfaces such as Collections, Recovery Management, and Treasury Optimizer etc. ACBS enhancements and new interface system are validated through system and integrated testing.

BI Reports (Datawarehouse Testing): Business object is a reporting tool. This project involves data validation and user interface testing of reports, generated by the BI development team using SAP Business Objects Product suites – Web Intelligence and crystal reports on different database (Such as Oracle, MYSQL and DB2 etc).

SASNPV: SASNPV is client based application based on SAS optimizing net present value with components such as LOB (line of business), End user, and Mail Decision and Aggregation tools. This project involves functionality and GUI testing of components in SASNPV.

Capstone: Capstone is a UNIX based credit decision engine. Capstone is a commercial off the shelf (COTS) application that Capital One has highly customized. This project involves testing of new / enhancement functionality in Capstone application.

Ameriprise Financial, Minneapolis, MN (Aug'07 - Aug'08)

Fixed Income: Fixed Income is web based application used to trade equity shares, options, futures and dept (Capital Markets). Understanding business knowledge of functionalities and technologies.This project involves testing of new enhancement functionality, system, integrated, regression and supporting User acceptance testing.

ABNAMRO Service Inc. Chicago, IL (Oct'05 - Jul '07)

RTS System Conversion, Automation and Testing: RTS Books and Records (RTS B&R), back office accounting system that complements Tradedesk, the front office trade capture platform. As a part of this project, replacing current Domestic Treasury back office system (InTrader) with an application (RTS Books and Records) that seamlessly integrates with Front Office system (TradeDesk)

Mosaic Teller: As a part of this project, testing Mosaic Teller Client/Server front end online banking application used for transactions, like, opening and maintaining Direct deposit account, opening and maintaining new saving account, Loan, Mortgage, Retirement, Transfer etc

Identity theft: As a part of this project, testing identity theft web based online application, offers an option to purchase an identity theft insurance product through a third party vendor.

FxPay: Testing of FxPay web based application enables users to execute and settle foreign exchange transactions in a fully interactive, straight through processing environment. This project involves user acceptance and system testing

State Street Bank, Mysore, India (Sep'03 -Sep'05)

SSB PRICING: The project involves User acceptance testing of 'Navigator'. 'Navigator' is an application used by Global Pricing Service (GPS) department of State Street to price mutual funds and calculate NAV (Net Asset Value) of Mutual funds. It is also used for generating mutual fund analysis reports. The project dealt with

SSB GCAS 3.0 and GCAS 2.0: This project involved system appreciation and User acceptance testing of 'Global Corporate Action System (GCAS)'. GCAS application automates Capture, Create, Build and Distribute notifications process involved in providing custodian services.

Responsibilities:

- Analyze business and system requirements. Involve in requirement walkthrough with client and business analyst to understand the scope and requirements.
- Prepare test plan which covers project essential requisites, test approach and strategy, testing scope, roles and responsibilities, testing timelines, entry and exit criteria and test schedule etc.
- Create Business risk Assurance Strategy (BAS) document based on business requirements which covers data strategy and risk mitigation.
- Estimate timeline and scheduled system testing to ensure timely and quality of deliverables.
- Allocated tasks along with managed / mentored up to 10 member team.
- Design test cases for testing based on BAS, Business and System Requirements.
- Created requirement traceability matrix through Quality Center for mapping test cases with requirements.

- Track defects through Quality Centre 10.0. Link defects with test cases and requirements.
- Complex test data creation as input data has to hit each and every rule to satisfy all the scenarios.
- Ensure all defects are closed before production rollout, and documenting defects in the system testing sign off document.
- Designed and developed Automation scripts using QTP in accomplishing regression testing.
- Utilized SoapUI and groovy scripts to automate web services.
- Keyword, Data Driven and Hybrid framework and Descriptive programming used to automate regression test cases.
- Experienced in writing SQL of ALL JOINS, UNION and Sub queries.
- Applied UNIX commands to run and compare output files of UNIX scripts.
- Automated web-based application using Selenium Webdriver and used TestNG framework
- Ensured all impacted downstream applications, due to changes in capstone application, are informed.
- Simultaneously tested multiple new / enhancement projects adhere to tight timelines.
- Understanding business knowledge of functionalities and technologies.
- Ran JCL jobs to validate backend data testing.
- Review test cases and test results done by testing team.
- Coordinated offshore calls to get ongoing project status update.
- Daily status call with clients to provide project status.
- Provide support to clients for User Acceptance testing, as well as, Go Live.
- Daily Sanity testing to ensure QA environment is working as expected.
- Ensured quality and timely delivery of all important testing documents.
- Confirmed all transaction data converted from Domestic treasury to RTS back office through UNIX shell Scripts and SQL Queries.

Environment:
HP ALM, HP QTP, SOAPUI, Selenium Webdriver, SQL developer, Teradata SQL Assistant, Oracle 10g, UNIX, SAP Business Objects Product suites – Web Intelligence, Crystal Reports, Teradata, HP Quality Center 10.0, TOAD, XMLspy, SAS language, SAS Datasets, MY SQL, DB2, Gupta SQL, JAVA, CORBA, AS400, Hummingbird- Exceed, Rapid SQL, Xrunner .

Education

Bachelor of Engineering in Electrical and Electronics Engineering
SRI Sivasubramaniya Nadar College of Engineering (SSNCE) | Madras University
Kalavakkam, Tamil Nadu, India
Graduated 2003

Certifications

Certified Software Tester (CSTE)
Chartered Institute of Bankers in Scotland
NSE's Certification in Financial Markets (NCFM)

Languages

English
Tamil